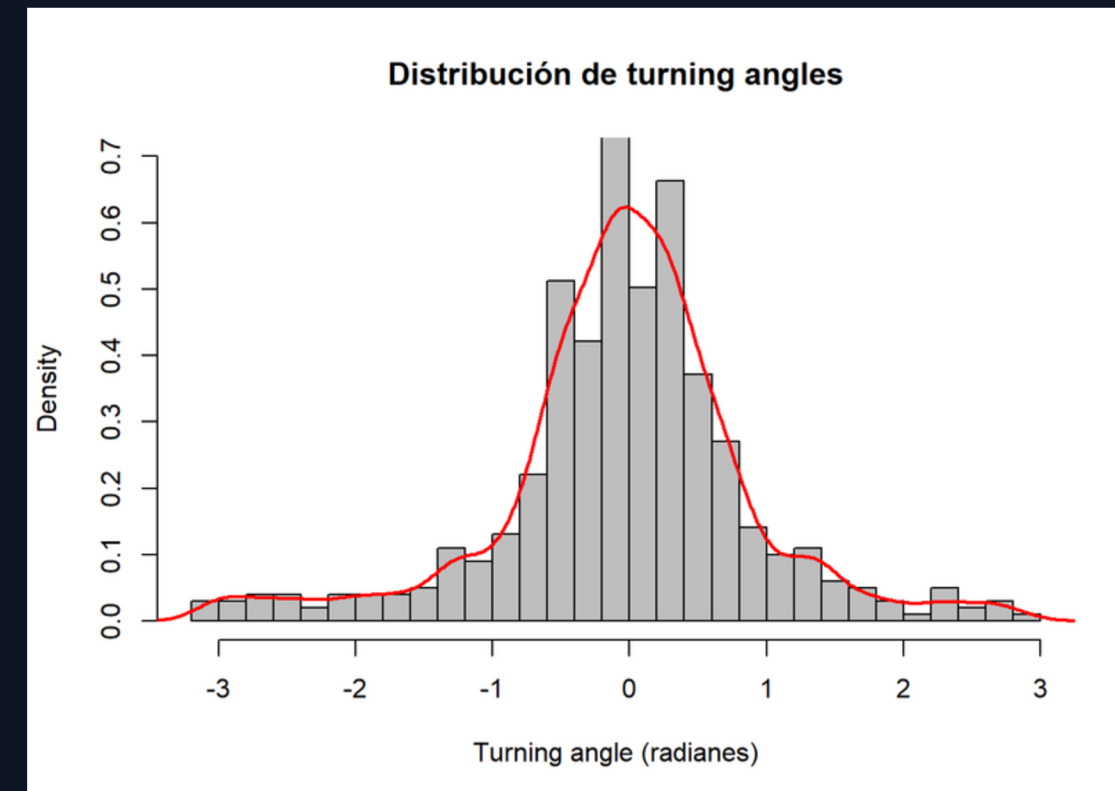
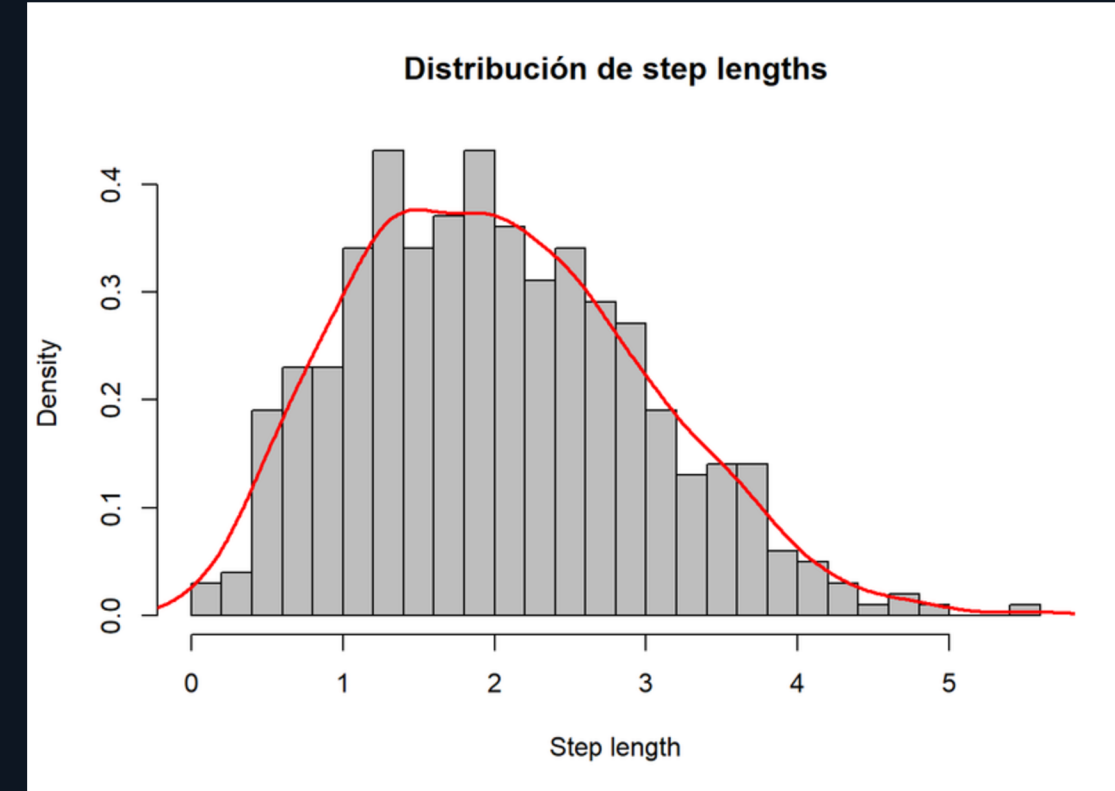
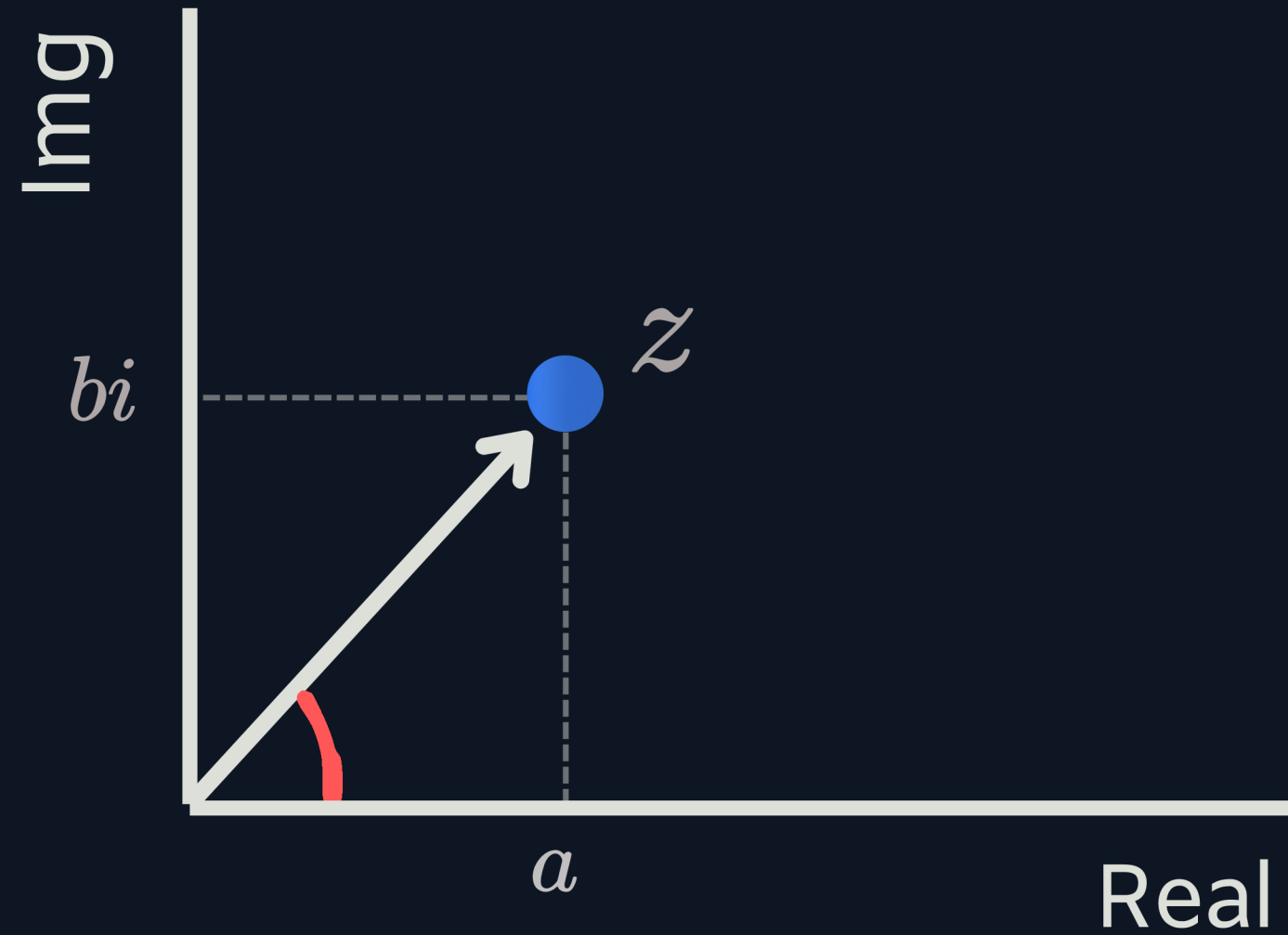


Métricas

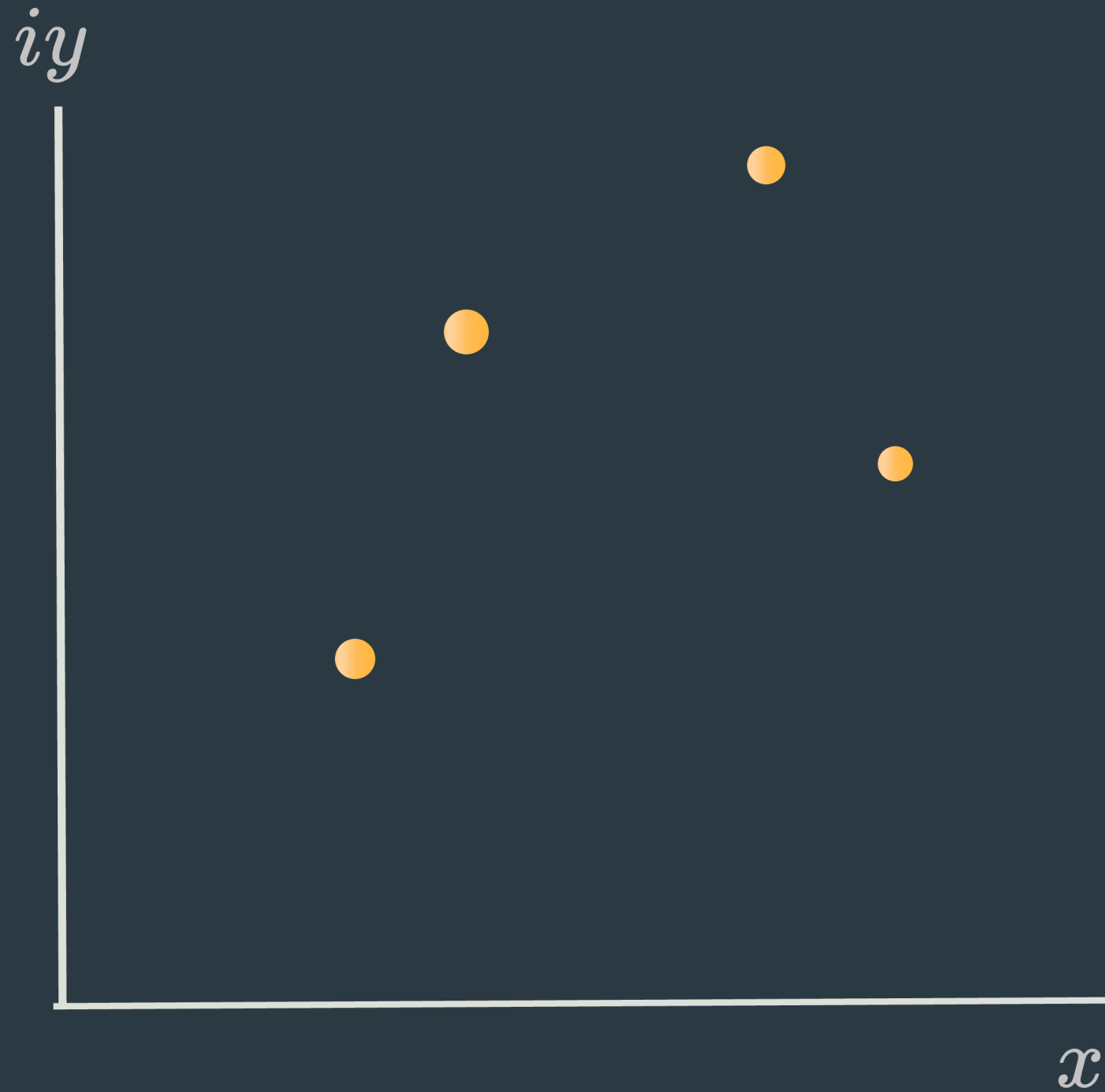
Medidas de resumen



Unidades básicas

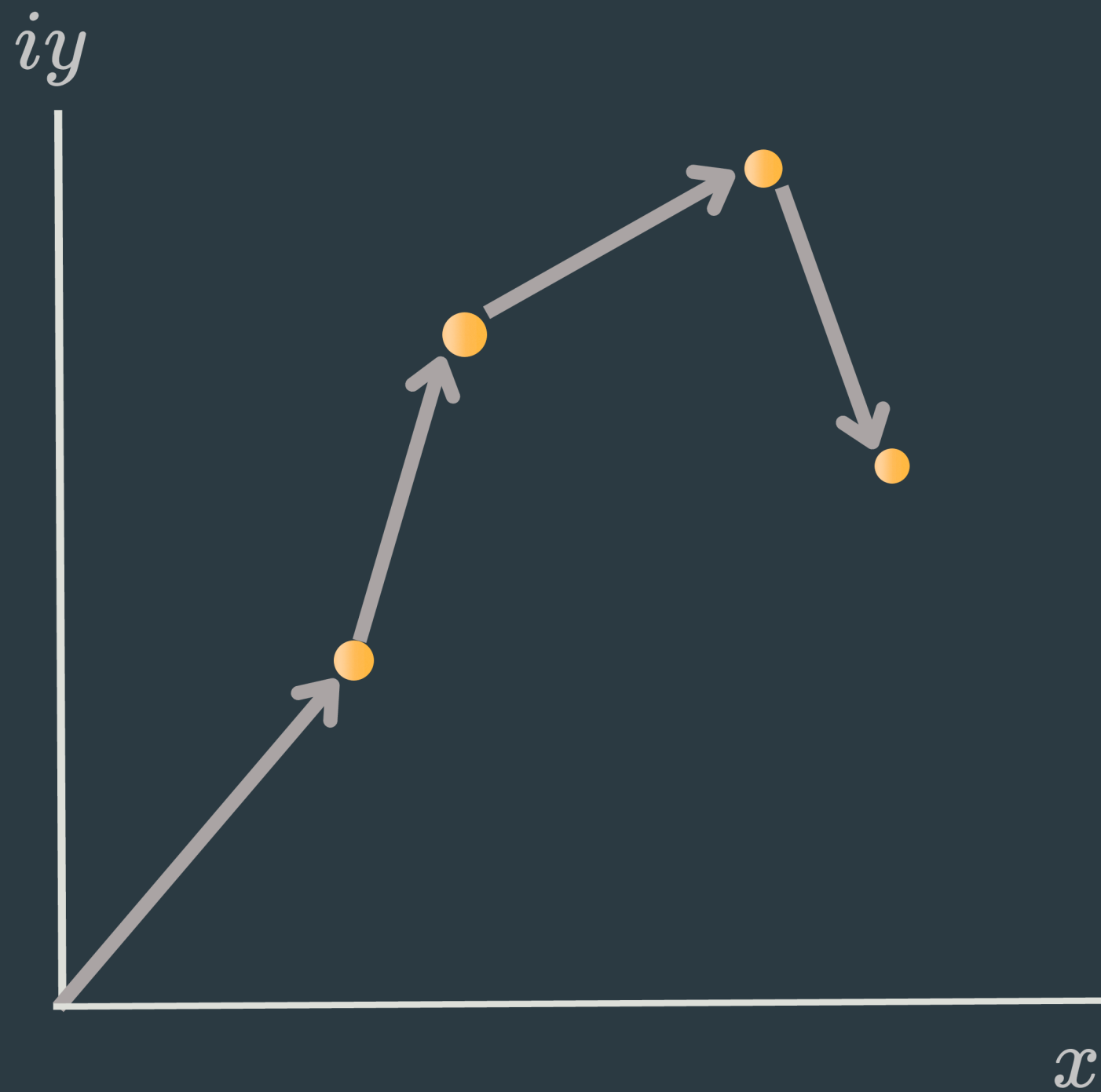
A nivel del paso

$$z_t = x_t + iy_t$$



Unidades básicas

A nivel del paso



1°

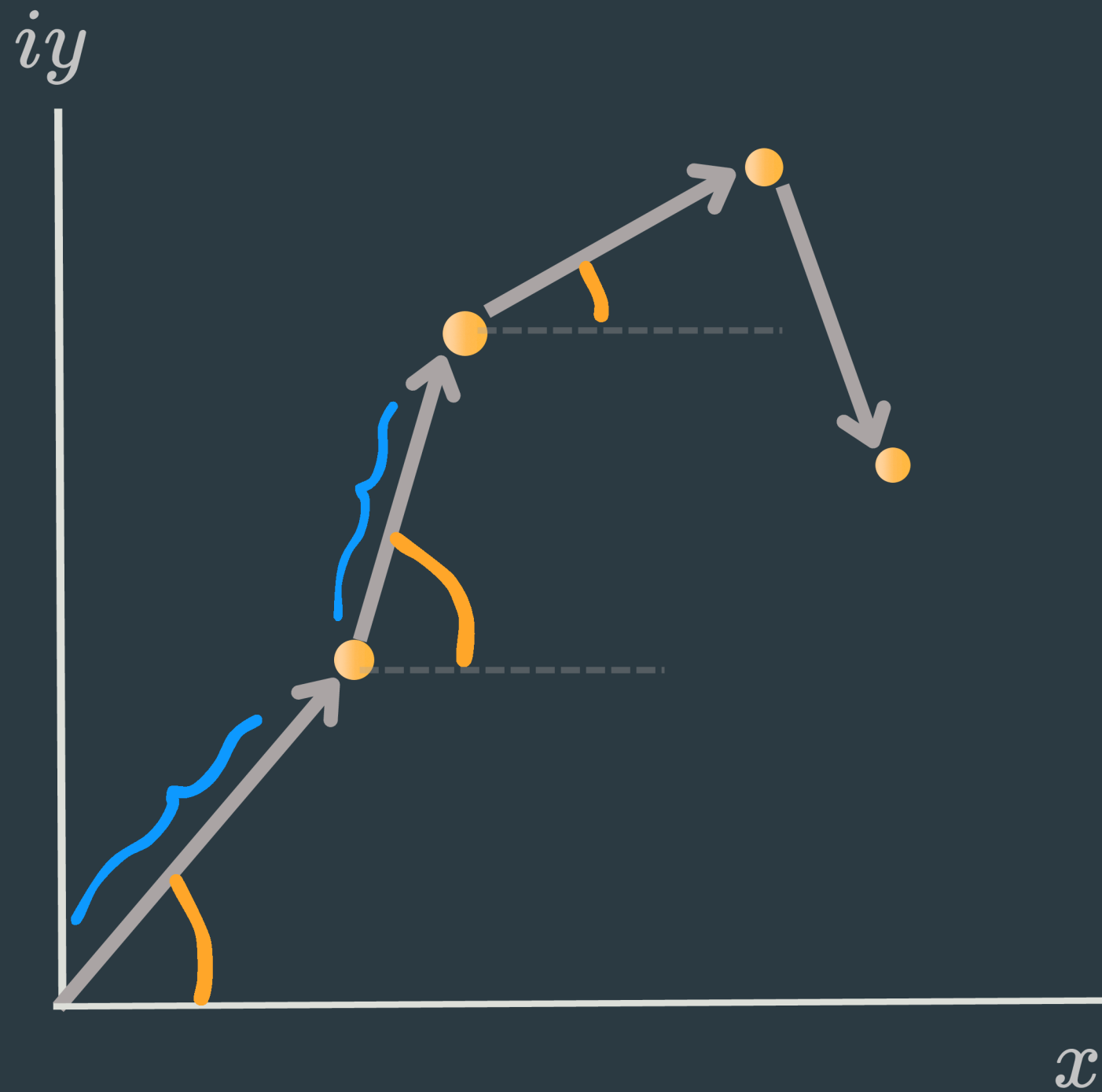
$$z_t = x_t + iy_t$$



$$\Delta z_t = z_{t+1} - z_t$$

Unidades básicas

A nivel del paso



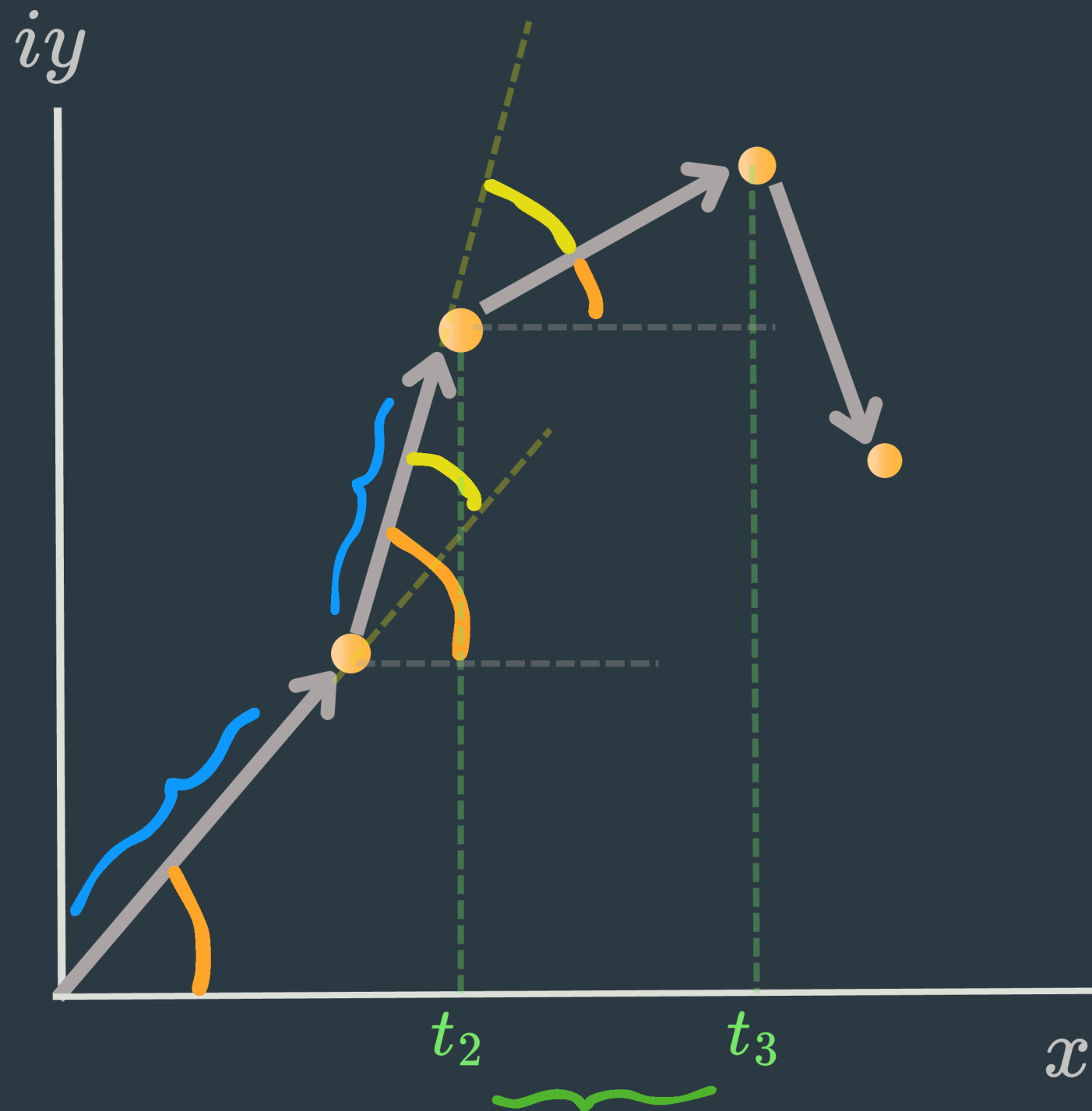
$$z_t = x_t + iy_t$$

$$\Delta z_t = z_{t+1} - z_t$$

$$\underline{L}_t = |\Delta z_t| \quad \underline{\theta}_t = \arg(\Delta z_t)$$

Unidades básicas

A nivel del paso



$$z_t = x_t + iy_t$$

$$\Delta z_t = z_{t+1} - z_t$$

$$\underline{L}_t = |\Delta z_t|$$

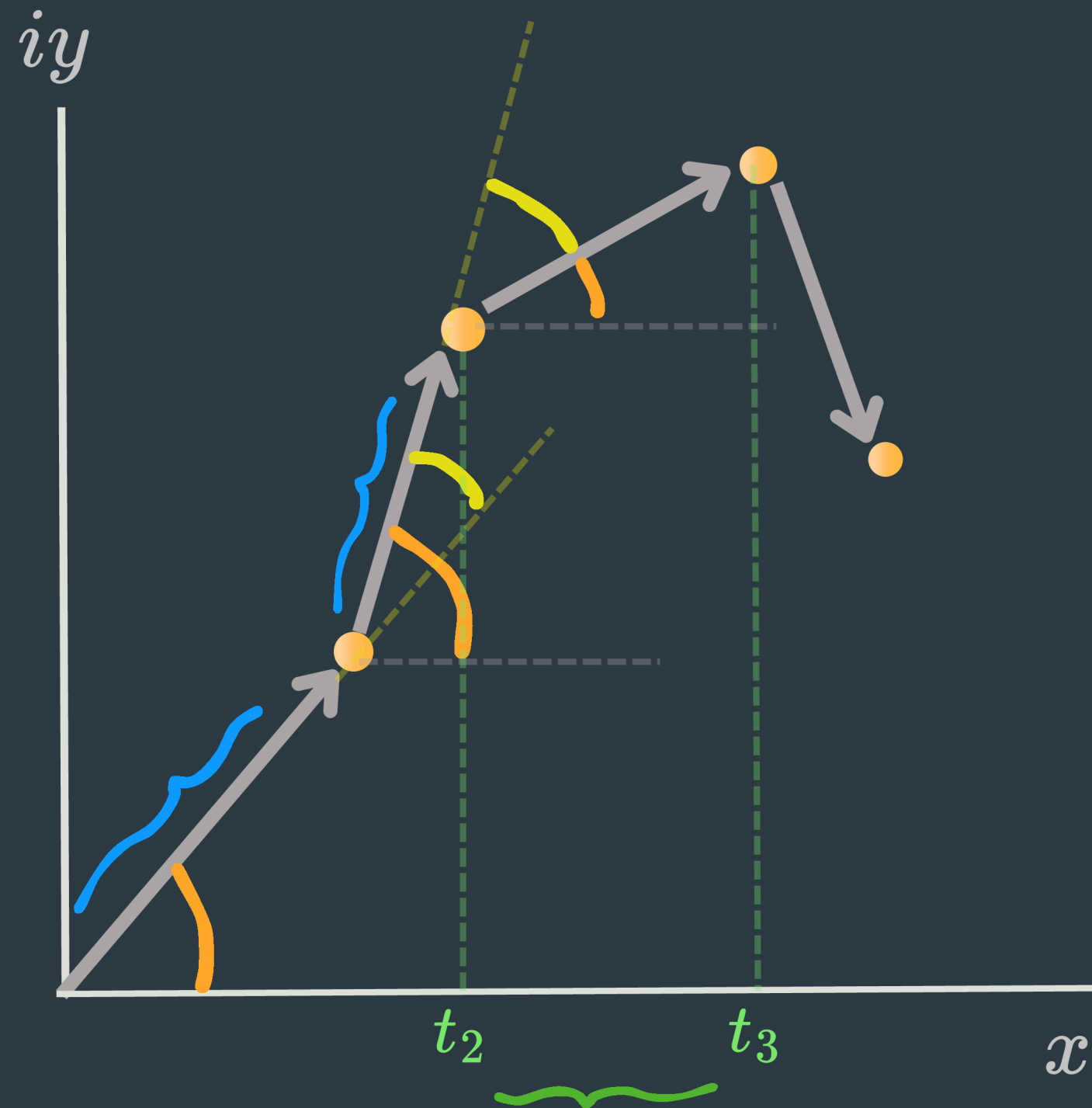
$$\underline{\theta}_t = \arg(\Delta z_t)$$

$$\underline{\Delta t}_t = t_{t+1} - t_t$$

$$\underline{\alpha}_t = \theta_{t+1} - \theta_t$$

Unidades básicas

A nivel del paso



$$z_t = x_t + iy_t$$

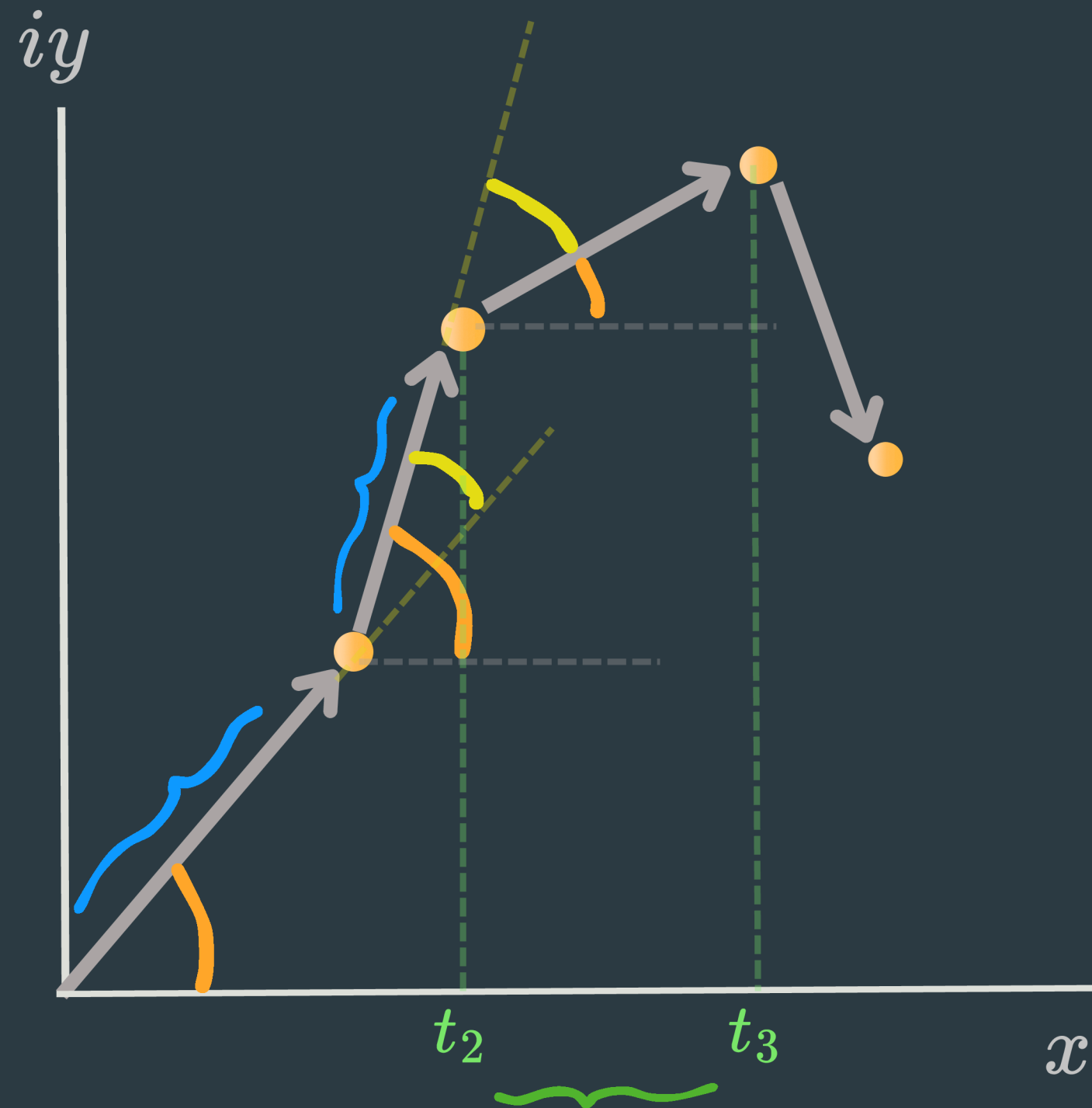
$$\Delta z_t = z_{t+1} - z_t$$

$$2^\circ \quad \underline{L}_t = |\Delta z_t| \quad \underline{\theta}_t = \arg(\Delta z_t)$$

$$\underline{\Delta t}_t = t_{t+1} - t_t \quad \underline{\alpha}_t = \theta_{t+1} - \theta_t$$

Unidades básicas

A nivel del paso



$$z_t = x_t + iy_t$$

$$\Delta z_t = z_{t+1} - z_t$$

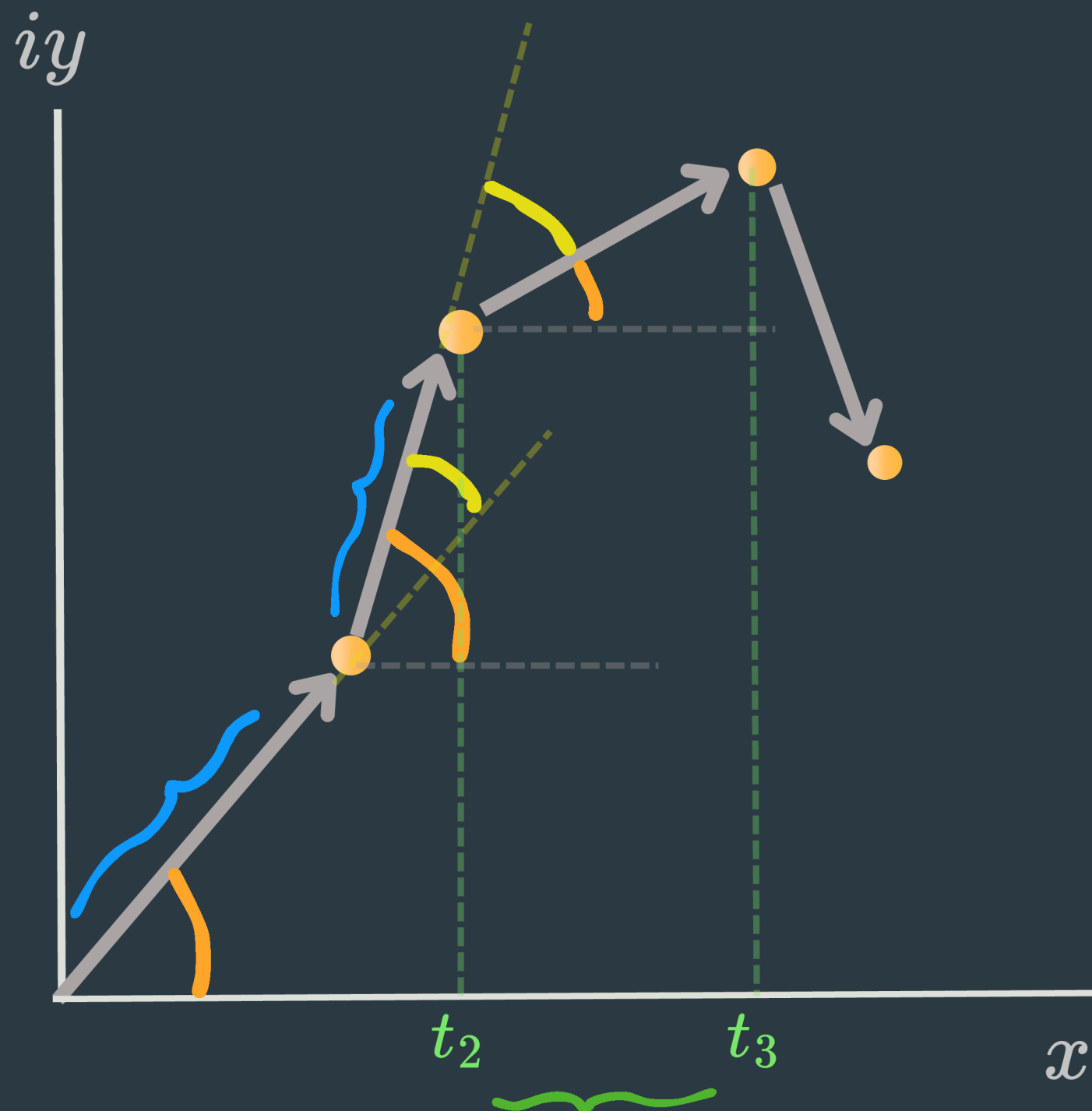
$$\underline{L}_t = |\Delta z_t| \quad \underline{\theta}_t = \arg(\Delta z_t)$$

$$\underline{\Delta t}_t = t_{t+1} - t_t \quad \underline{\alpha}_t = \theta_{t+1} - \theta_t$$

$$\underline{v}_t = \frac{L_t}{\Delta t_t} \quad \underline{v_i}(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta L}{\Delta t}$$

Unidades básicas

A nivel del paso



$$z_t = x_t + iy_t$$

$$\Delta z_t = z_{t+1} - z_t$$

$$\underline{L}_t = |\Delta z_t| \quad \underline{\theta}_t = \arg(\Delta z_t)$$

$$\underline{\Delta t}_t = t_{t+1} - t_t \quad \underline{\alpha}_t = \theta_{t+1} - \theta_t$$

$$3^\circ \quad \underline{v}_t = \frac{L_t}{\Delta t_t} \quad \underline{v_i}(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta L}{\Delta t}$$

Métricas Globales – Distancias

$$d_t = |z_t - z_1|$$

¿Qué tan lejos está el animal de su punto de partida en cada momento?

$$D_{\text{neta}} = |z_n - z_1|$$

¿Cuánto se alejó en todo el muestreo?

$$MDD = \max_t d_t$$

¿Cuál fue la máxima distancia alcanzada desde el punto inicial?

$$D_{\text{total}} = \sum_{t=1}^{n-1} |z_{t+1} - z_t|$$

¿Cuánto camino acumulado recorrió el animal?

$$NDS_t = |z_t - z_1|^2$$

¿Cómo cambia este desplazamiento en el tiempo?

$$MSD(\tau) = \frac{1}{n - \tau} \sum_{t=1}^{n-\tau} |z_{t+\tau} - z_t|^2$$

¿Cuánto se separa, en promedio, un individuo de sí mismo después de un intervalo de tiempo τ ?

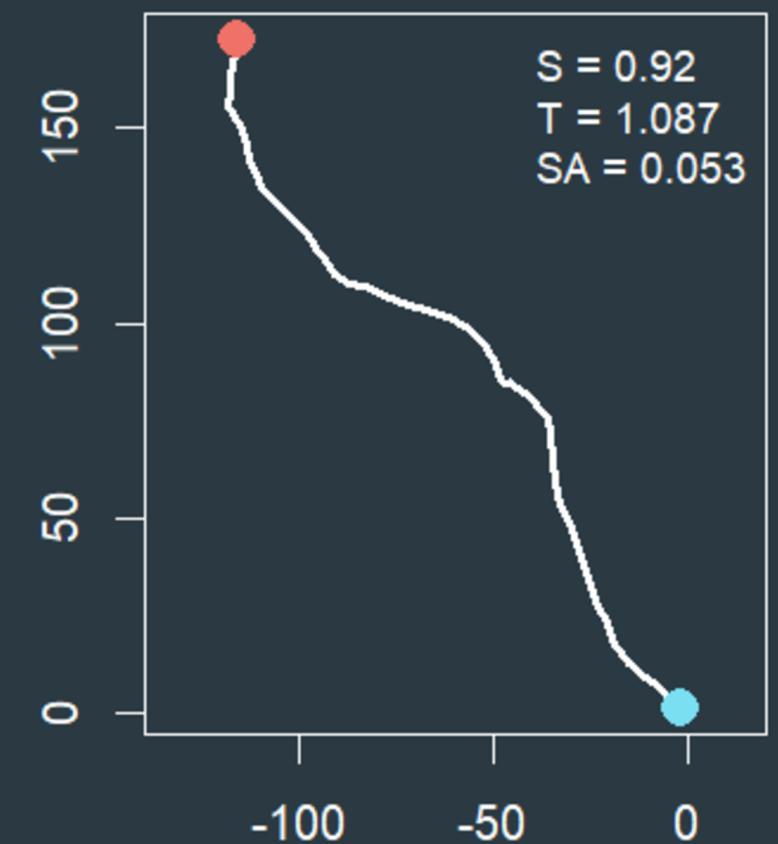
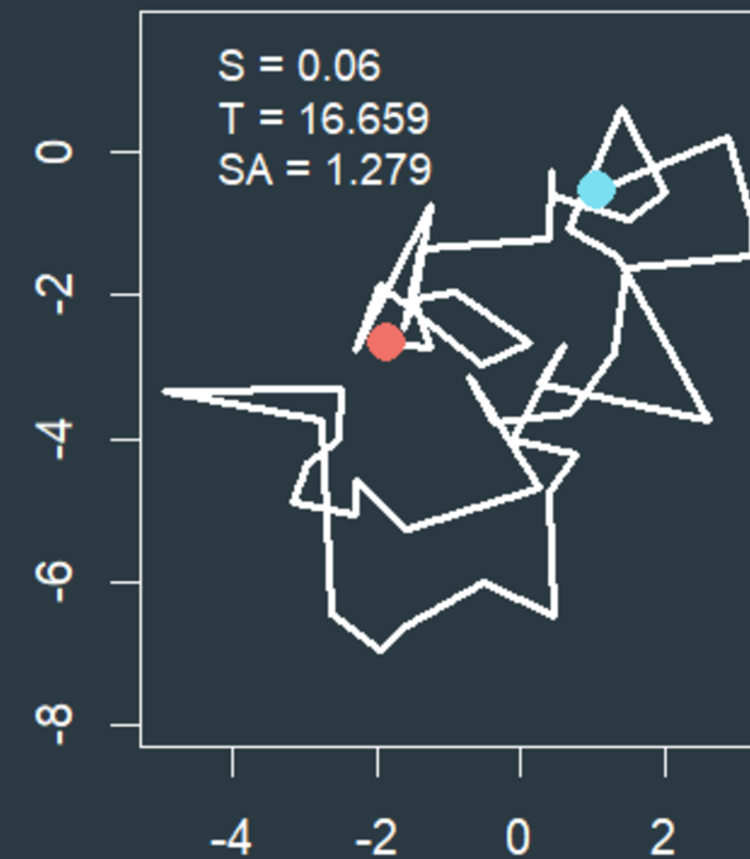
$$\tau = lag$$

Métricas Globales – Ángulos

$$S = \frac{D_{\text{neta}}}{D_{\text{total}}} \quad \text{Straightness}$$

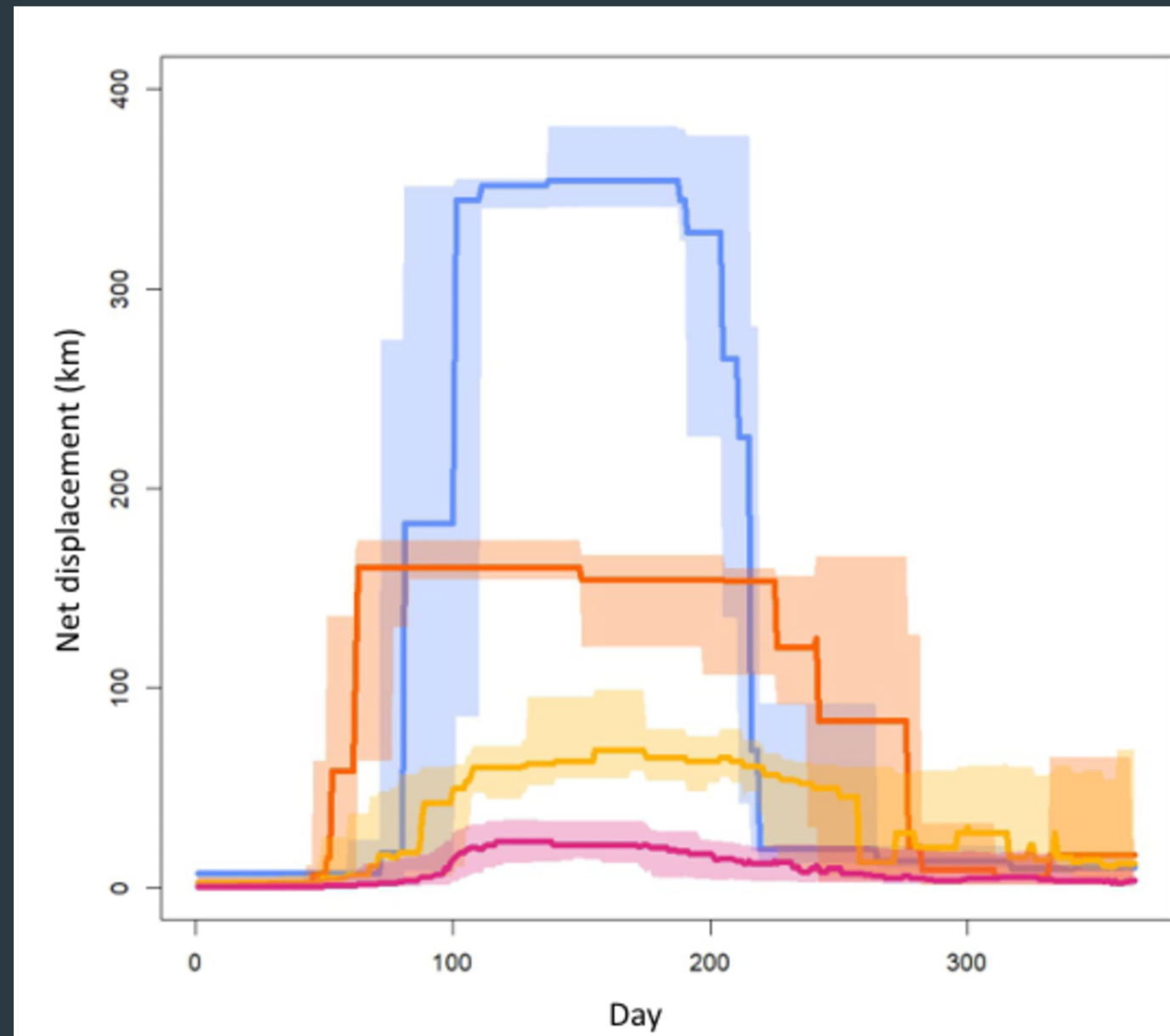
$$T = \frac{D_{\text{total}}}{D_{\text{neta}}} \quad \text{Tortuosidad}$$

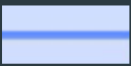
$$SA = \frac{\sum_{t=1}^{n-2} |\alpha_t|}{D_{\text{total}}} \quad \text{Sinuosidad angular}$$



Migración o fidelidad de sitio

NDS



-  Long-distance migrant
-  Medium-distance migrant
-  Shorth-distance migrant
-  Resident

Caracterización del movimiento de Orcas



Candidate mode	NSD curve	Equation	Path example
Resident		$NSD = c$	
Nomad		$NSD = a \cdot t$	
Round trip		$NSD = \frac{A}{1 + m \cdot \exp(-k(t - i))^{1/m}} + \frac{A'}{1 + \exp(-k'(t - i'))}$	
Partial trip		$NSD = \frac{A}{1 + \exp(-k(t - i))} + \frac{A'}{1 + \exp(-k'(t - i'))}$	
Transit		$NSD = \frac{A}{1 + \exp(-k(t - i))}$	

$$NSD = \frac{A}{1 + m \cdot \exp(-k(t - i))^{1/m}} + \frac{A'}{1 + \exp(-k'(t - i'))}$$

